

KANFeel: A Novel Kolmogorov-Arnold Network-Based Multimodal Emotion Recognition Framework

Le Fang

School of Design
The Hong Kong Polytechnic University
Hong Kong SAR, China
leonardo-le.fang@connect.polyu.hk

Yingqing Xu

The Future Laboratory
Tsinghua University
Beijing, China
yqxu@tsinghua.edu.cn

Bo Chai

School of Design
The Hong Kong Polytechnic University
Hong Kong SAR, China
bo-chris.chai@polyu.edu.hk

Stephen Jia Wang

School of Design and Research Centre for Future (Caring)
Mobility
The Hong Kong Polytechnic University
Hong Kong SAR, China
stephen.j.wang@polyu.edu.hk

Abstract

Multimodal emotion recognition (MER) has become a significant interdisciplinary research area in HCI by analyzing various data modalities. Recent studies have shown promising results in MER by fusing features across modalities. However, efficiency in real-world scenarios remains a challenge. To improve the model efficiency, Kolmogorov-Arnold Networks (KANs) have been proposed as efficient alternatives to Multi-Layer Perceptrons (MLPs). However, employing KANs in MER is still underexplored. This paper introduces **KANFeel**, a novel KAN-based MER framework that processes multimodal inputs to predict emotional states. Furthermore, we adopt the KAN-based model to replace the attention mechanism in the transformer, referred to as **KANFeel-Attent**, to achieve enhanced recognition performance. Comprehensive evaluations across three public datasets are conducted to analyze the efficiency improvements of KANFeel by comparing model parameters and training speeds. Finally, emotion recognition enhancements, measured through accuracy and F1-score, are validated using the KANFeel and KANFeel-Attent models against baseline and state-of-the-art methods.

CCS Concepts

• **Applied computing** → **Health care information systems**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Multimodal Emotion Recognition, Kolmogorov-Arnold Networks, Deep Learning, Human-Computer Interaction

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1 Introduction

As a cross-disciplinary study for human beings, emotion recognition (ER) has kept growing as a significant research direction, spanning researchers from academic fields like Artificial Intelligence, Healthcare, Interaction Design, etc. According to the advancements of AI technologies through diverse modalities, multimodal emotion recognition (MER) has become a prominent research topic [11, 12, 22, 40] in HCI field. Empowered by various interactive ways between humans and devices, multiple modalities can be extracted and analyzed for predicting humans' current emotional state, including audio signals and visual expressions [22, 40], wearables' sensory data and human behaviors [11, 12].

Recent years have seen a growing trend for MER tasks based on deep learning methods. Li et al. [22] introduced an outstanding CFN-ESA model for emotion detection in conversations by fusing features across textual, visual, and acoustic modalities. Tu et al. [40] integrated acoustic, visual, and contextual modalities for more reliable conversational emotion recognition. Besides, recently various deep learning architectures were proposed (such as the autoencoder-based model [12] and the long short-term memory (LSTM) based framework [11]) with outstanding detection results based on electroencephalogram (EEG) signals and human behaviors.

Despite the recent advancements in model performance for MER tasks, efficient deep learning [27] remains challenging in real-world environments, due to the growing large deep learning models and heavy training costs. To enhance the efficiency when training AI models in MER, researchers intend to reduce the computational costs by extracting more significant features [17, 26] and adopting

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a more efficient model architecture [4]. Recently, the Kolmogorov-Arnold Network (KAN) [24] was proposed and verified as an efficient and effective alternative to Multi-Layer Perceptron (MLP) from the mathematical perspective. Besides, most recent studies have verified the improved training efficiency of KAN-based deployments for deep learning tasks [23] and the higher generalization ability by integrating with other advanced models (such as convolutional, recurrent models) [38]. However, there still exists a research gap in exploring the potential of applying KANs for MER tasks.

To address the research gap, this paper proposes a novel KAN-based MER framework: **KANFeel**, which adopts multimodal sequences as input and predicts the emotional state more efficiently. Besides, to explore the generalization ability of the KAN-based model with other advanced models in MER tasks, we further integrated the transformer-based model [41] and KANFeel, namely **KANFeel-Attent**, by replacing the attention mechanism for both improved training efficiency and model effectiveness. To evaluate the efficiency of the novel framework, both model parameters and speed (training and testing phases) are evaluated and compared with other baseline models across three multimodal emotion datasets. Further experimental results are conducted to verify the effectiveness of **KANFeel** and **KANFeel-Attent** by comparing with other baselines and state-of-the-art (SOTA) methods across three public datasets.

2 Related Works

2.1 Comparison of Unimodality and Multimodality for Emotion Recognition

Emotion recognition tasks intend to reply on various human data to identify and recognize humans' emotional states effectively, such as facial expressions [35], speech samples [13], behaviors [10], text information [7], and physiological signals [11]. For unimodal emotion recognition, it adopted a single modality and directly predicted the emotional signs through the emotional classifier. For example, facial expressions [35] and speech signals [13] are common human information adopted for training deep learning models in emotion monitoring tasks. Recently, EmoSense [10] adopted human gesture features to predict stress levels through the microgesture touch sensor.

However, relying on only one modality brings more limitations, lower robustness, and poorer effectiveness in complex real-world scenarios, while multimodal data can address potential concerns from three benefits. First, accessing various modalities that may depict the same emotional state (like we might smile and appaude when we are happy) can show more contextual information and provide more robust recognition results [5]. Second, multimodal inputs typically complement each other and address their respective shortcomings [44]. Third, the emotion recognition task can be completed through other data inputs by compensating for the missing input from a specific modality [9], like a user is not speaking (audio input) but smiling (visual input).

Recently, empowered by AI and sensory developments, multimodal emotion monitoring (MER) has gained more attention to be a prominent research direction in the HCI field [21]. Apart from the unimodal method, MER intends to extract and fuse features from multiple sources and predict human emotion more effectively and

robustly in complex scenarios through various interactive methods. For example, Tu et al. [40] introduced a more reliable CMERC structure by integrating acoustic, visual, and contextual modalities to track emotional states in conversations. Besides, EmoMG-framework [11] was proposed to employ multichannel EEG signals and gesture features through sensory devices, recognizing humans' emotional changes based on the LSTM network with state-of-the-art (SOTA) performances.

2.2 Multimodal HCI for Emotion Recognition

Recent years have seen advancing developments in multimodal emotion recognition. Unlike unimodal emotion recognition, MER intends to preprocess features from various data sources, distill the potential features conveying emotional features, and fuse them for more effective emotion recognition [5]. Multiple modalities can be accessed through various interactive methods and devices, commonly like cameras [3] for visual inputs, microphones [3] for audio inputs, wearables (like EEG headsets [11]) for recording physiological data, mobile phones [39] for tracking more hybrid and contextual information, and other sensory devices [10] for tracking human behaviors. After acquiring the raw data through multiple devices, potential outliers from separate data samples should be first identified and removed for data normalization. After the data cleaning phase, data features should be extracted through various techniques, like convolutional neural networks (CNNs) for visual data and recurrent neural networks (RNNs) for temporal data [5]. Due to the inconsistency among various modalities, feature fusion still remains a big challenge. In general, feature fusion [5] can be made at the early stage when combining the features before importing them for the recognition model or at the late stage by merging and analyzing the recognition results through individual modalities. Besides, a hybrid fusion method can also be involved to integrate the strength of early fusion and late fusion stages. After conducting the feature fusion, traditional ML and deep learning models can be employed for training the emotion classifier based on the features and the corresponding emotional labels (like the Valence-Arousal-Dominance ratings [6] or discrete emotional states [15]). Traditional ML models, like Random Forest, KNN, and Decision Tree, have been verified with great emotion recognition performances in recent studies [11] for small datasets. Advancing deep learning models are also verified with SOTA recognition performances, empowered by LSTM [11], GNN [48], and Transformer [36].

For practical use scenarios in HCI, multimodal approaches have been increasingly integrated into various applications. For example, an effective emotion-aware chatbot was developed by incorporating text, audio, and video cues to enhance the user experience [21]. Moreover, the advent of wearable and mobile technologies enables the use of wearable signals to improve emotion recognition performance in real-world applications [8]. Besides, the audio smartwatch voice messaging system [1] was recently proposed to enhance emotional communication by incorporating emotionally animated teasers. Additionally, the integration of biosensors and speech data within virtual reality (VR) environments [42] provides more objective measures for assessing participants' stress levels and emotional states. Gamification techniques are also gaining traction,

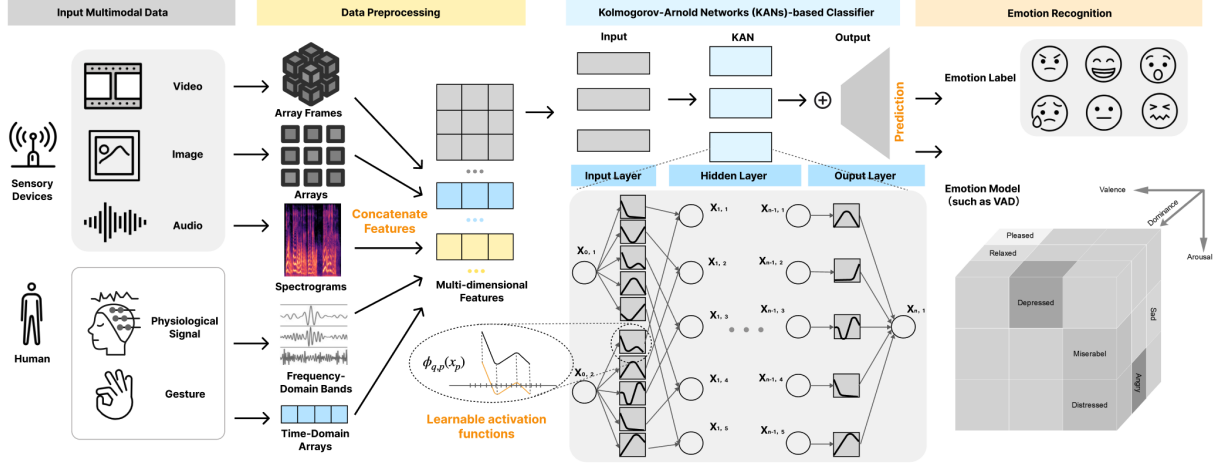


Figure 1: Framework for Multimodal Emotion Recognition Using Kolmogorov-Arnold Networks (KANs)

particularly in improving facial emotion recognition accuracy and increasing user engagement [47]. Furthermore, the collection and analysis of multiple physiological data streams have been shown to reveal various affective states during VR-based exergaming experiences [32].

2.3 Efficiency Improvements for Training Deep Learning Models in Multimodal Emotion Recognition

With the growing amount of adopted data, training deep learning models for MER tasks has brought increasing computational costs as well. To provide more efficient model training strategies, recent studies also explored potential solutions across various use scenarios for emotion recognition from data-centred and model-centred efficient training perspectives [34].

Data-centric strategies intend to reduce the computational complexity of the data samples and improve the training efficiency through data augmentation. Kaya and Karpov [17] proposed the computationally efficient feature normalization strategies by deploying a cascaded normalization approach to fuse features for training ML models for cross-corpus and cross-language emotion recognition. Ma et al. [26] proposed the cross-modal noise modeling for the audio and video data to eliminate most of the data pollution in the audio channel and the data redundancy in the visual channel. Wu et al. [43] recently verified the enhanced efficiency of adopting olfactory to evoke users' EEG responses for emotion recognition model training toward the video-based stimuli.

Model-centered strategies focus more on the computational complexity of adopted AI models via efficient model architectural choices. Satt et al. [33] proposed the deep network comprising convolutional and recurrent networks to learn emotion-aware information more efficiently from speech spectrums. Bai et al. [4] adopted lightweight binary tiny machine learning (TinyML) and tiny deep learning models for efficient emotion recognition on portable edge devices with reduced parameters and computational needs.

3 Proposed Method

This study employs KANs for a novel multimodal emotion recognition framework, as illustrated in Fig. 1. Through various data modalities, the features are extracted and fused through the concatenation process. Then fused features are input into a KAN-based classifier model equipped with learnable activation functions. Finally, the classifier calculates the probability distribution of various emotions, providing an emotion prediction mapping to a specific emotion label.

3.1 Data Preprocessing and Feature Fusion

Considering the diversity of input data sources, multiple techniques would be adopted to transform input data into various representations. For “text” information, vectorization through Word2Vec [28] can be adopted to make contextual information into word embeddings. For “audio” information, feature extraction is done by computing the Mel-frequency cepstral coefficients (MFCCs) through Mel Spectrograms [31]. For continuous “sensor signals”, the Fourier transform can be applied to convert them into various bands features [11] for the following deep learning task. For “visual data” (like images, videos, etc.), photos (images can be represented as multiple image frames) can be extracted as normalized arrays through the OpenCV library [14] and VGG16 (a pre-trained CNN model) [37].

The feature fusion intends to integrate multimodal inputs under a single time interval into a single input presentation. To ensure all features are on the same scale, Min-Max scaling or Z-score normalization can be adopted for feature normalization. Then various normalized features are concatenated into a single feature vector and PCA (Principal Component Analysis) [18] can be applied to reduce the vector dimensionality in case of the overly large vector.

3.2 KAN-based Classifier

As an efficient alternative to the MLP with fewer parameters, we replace the original MLP layers with KANs to construct the classifier. KANs rely on the Kolmogorov-Arnold representation theorems

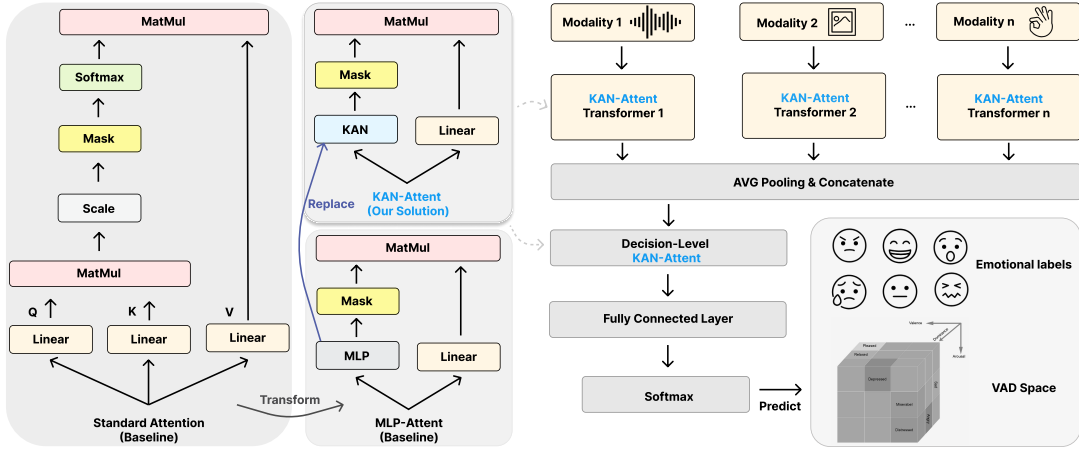


Figure 2: KANFeel-Attent: The Framework for Multimodal Emotion Recognition based on KAN-based Attention Mechanism

[19], which represent continuous functions with a sum of simple functions. A KAN can be formed by a multivariate function $f(x)$ [24]:

$$f(x) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{p,q}(x_p) \right), \quad (1)$$

in which $\Phi_q, \phi_{p,q}$ are univariate functions that map each input x_p , such as $\phi_{p,q} : [0, 1] \rightarrow \mathbb{R}$ and $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$. Note that $\phi_{p,q}(x_p)$ is a learnable activation function. According to the Kolmogorov-Arnold representation theorem [19], the inner functions construct a KAN layer with $n_{in} = n$ and $n_{out} = 2n + 1$, and the outer functions make another KAN layer with $n_{in} = 2n + 1$ and $n_{out} = n$. So the KAN representation contains two KAN layers in this case. To approximate univariate functions well, the 3-order B-splines were approximated by the radial basis functions (RBFs) with Gaussian kernels [23]. Then KAN can be reformed as:

$$f(x) = \sum_{p=1}^n w_p \phi(\|x_p - c_p\|), \quad (2)$$

where w_p are adjustable coefficients and ϕ is the RBF based on the Gaussian kernel to measure the distance between the input x_p and the center c_p . $\phi(r)$ is defined as $\phi(r) = \exp(-\frac{r^2}{2h^2})$, where $r = \|x_p - c_p\|$ is the radial distance and h is a parameter that controls the spread of the function.

We further define the **learning task** to recognize the emotional state based on the multimodal input. For various input representations, the input layer of the KAN-based classifier contains the fused input feature from multiple modalities. The hidden layers extract the deeper representations through KANs with the above learnable activation function $f(x)$. To prevent the model from overfitting, the dropout layer was added with the dropout rate of 0.5. The normalization layer was used to accelerate the model training. Finally, a softmax layer is adopted to predict the probability distribution over various emotional labels. Considering the multiclass recognition task, we define the loss function as the Categorical Cross-Entropy Loss $\mathcal{L}_p$ considering a single sample x_p :

$$\mathcal{L}_p = - \sum_{i=1}^C y_{p,i} \log(\hat{y}_{p,i}), \quad (3)$$

where C means the number of emotion classes. For the input sample x_p , $y_{p,i}$ indicates the true label of the one-hot encoded vector (1 for true and 0 for other classes) and $\hat{y}_{p,i}$ means the predicted probability for the j^{th} class. Note that $\mathcal{L}_p$ turns to the Binary Cross-Entropy Loss when $C = 2$.

3.3 Optimization of Attention Mechanism

To further improve the performance of the KAN-based model, we further replaced the attention mechanism [29] with the KAN-based architecture to augment the transformer model. As illustrated in Fig. 2, the original model [41] adopts the dot-product attention between query (Q) and key (K) vectors to compute attention weights in its transformer. To improve efficiency, a recent study [29] adopted the MLP layer to compute the attention weights. These weights are learned by the model and highlight the most important parts of the input. Inspired by [29], we replaced the MLP layer with KAN to compute the attention weights from the embeddings. Based on the proposed attention mechanism, we further deployed the KAN-Attent-based Transformer for MER tasks. In our scenario, the attention mechanisms across various stages are used to process (feature extraction phase) and combine (decision-level phase) multimodal data inputs. These attention weights determine which features of the input should be prioritized during processing and contribute the most to the final decision (emotional label). First, inputs from multiple modalities (audio, visual, text, etc.) are processed by KAN-Attent-based Transformers. Afterward, extracted embeddings are pooled based on the average pooling function (AVG Pooling) and then concatenated as the multimodal representation [49] for the further emotion recognition task. Then a decision-level KAN attentional mechanism further refines the predictions. Finally, outputs are mapped into either emotional labels (e.g., happy, sad, angry) or the VAD emotional space. We name this framework **KANFeel-Attent** by integrating KAN into the transformer models.

Table 1: Speed Performances Comparison of Models on Different Datasets (The bold values indicate the better results.)

Model	# of Params (N)	Emo-MG (ms)		# of Params (N)	IEMOCAP (ms)		# of Params (N)	EMOTIC (ms)	
		T_{train}	T_{test}		T_{train}	T_{test}		T_{train}	T_{test}
MLP	1,201	353.69	2.53	10,504	1843.08	33.14	2,519,001	268,582.48	1,049.11
KANFeel	473	166.62	7.84	4,253	974.55	55.41	1,057,949	27,491.92	1,124.27

4 Experiments

4.1 Datasets

This study adopts various multimodal datasets, including Emo-MG, IEMOCAP, and EMOTIC. Emo-MG [11] contains 790 samples from 15 participants labeled in Valence-Arousal-Dominance (VAD) emotional space, including 70 sets of EEG frequency bands (14 channels $\times$ 5 bands), and the frequency counts of micro gestures (MGs). IEMOCAP [6] contains approximately 12 hours of audiovisual data, including video, speech, and motions, to categorize various emotions. Since the emotional labels are unevenly distributed, four categories are selected with majority votes: anger, happiness (merged with excitement), sadness, and neutral [15]. For ease of model training, we adopted 7580 sets of audio signals and video frames based on the dialogue sentences. After the feature extraction and concatenation process, each sample length was 263,006 and reduced to 100 by adopting the PCA method [18]. We note the cumulative explained variance from the first 100 extracted dimensions is already more than 97.28%, indicating the higher efficiency of preserving most of the data variance with much fewer principal components. EMOTIC [20] is a database of 23,571 images and 34,320 annotated emotional states in VAD emotional space.

4.2 Baseline Models and Metrics

Since KAN was represented as the alternative to MLP architecture, we first adopted the MLP-based classifier as the baseline for comparison. Other common machine learning models were also trained for the recognition task, such as SVM, KNN, Random Forest (RF), AdaBoost, Naive Bayes Classifier (NBC), and SGDClassifier (SGD). Finally, various deep learning and state-of-the-art (SOTA) methods were selected for comparison for various datasets. For the Emo-MG dataset, we employed the Emo-MG (LSTM) framework [11], which leverages long short-term memory (LSTM) networks, along with the graph neural network (GNN) model called RGNN [50]. For the IEMOCAP dataset, we selected the most recent GNN-based models, CORRECT [30] and COGMEN [16], to conduct model comparisons. In the case of the EMOTIC dataset, our study compared with the Context-CNN based model EMOT-Net [20] and a GNN-CNN model [48]. For comparing model performances, *accuracy* and *F1-score* are adopted during the evaluation phase.

4.3 Implementation Details

All experiments were conducted on an Apple M2 Pro with 16 GB of memory (without additional GPUs), which also allows us to exclude some large models that require high computational costs in

the experiments. The training-to-testing ratio was set at 9:1, following a 10-fold cross-validation approach. To prevent overfitting, we set a maximum of 20 epochs and adopted the early stopping strategy (stopping when the validation set loss does not drop for 3 consecutive epochs). For the deep learning models, the batch size was set to 32, the learning rate to 0.001, and the Adam optimizer was used. To enhance the performance of the KAN-based layer with advanced models, the transformer-based model layers were incorporated, resulting in a new variant called KANFeel-Attent as mentioned in the prior section. As suggested by [49], there are 12 heads and 4096 feedforward sizes for each transformer layer.

5 Evaluation Results

5.1 Model Parameters and Speed

As an alternative to MLP, KAN has been credited with improving efficiency performance [23]. However, real-world MER performance has yet to be validated. Rather than relying on large computing devices or cloud services with powerful CPUs and GPUs, our work presented a pilot study conducted on a laptop with just 16 GB of memory and no additional GPU resources. To compare the KAN-based model with MLP, the number of model parameters and speed values are summarized in Table 1 for comparison. Besides, the speed is compared over these models as the average time costs on each dataset during the model training process (T_{train}) and the model testing phase (T_{test}).

On the *EMO-MG* dataset, the KAN-based model has significantly fewer parameters ($N = 473$) than the MLP classifier ($N = 1,201$) and a faster training phase, though its testing phase is slower, likely due to reshaping required for test samples. We further evaluated the efficiency of *EMO-MG-LSTM* on the *EMO-MG* dataset, finding that it has 40,901 parameters, with $T_{train} = 2131.36$ ms and $T_{test} = 134.04$ ms, indicating much lower efficiency across all three metrics. Similar trends are observed on the *IEMOCAP* and *EMOTIC* datasets. For *IEMOCAP*, KANFeel has fewer parameters (4,253) and faster training (974.55 ms), but slower testing (55.41 ms) compared to MLP (10,504 parameters, 1,843.08 ms training, 33.14 ms testing). On *EMOTIC*, the KAN-based model ($N = 1,057,947$) requires less than half the parameters of the MLP classifier ($N = 2,519,001$) and is nine times faster in training, with comparable performance.

In summary, these results align with the theoretical findings in [24], demonstrating the superior efficiency of the KAN-based model in MER tasks, particularly in terms of model parameters and training speed. Key insights across the datasets include: 1) The KAN-based model consistently requires fewer parameters and achieves faster training times. 2) The MLP-based model excels in testing times. 3) The test performance gap narrows for larger datasets.

Table 2: MER Performances of Models on Various Datasets (The “V”, “A”, “D” present the “Valence”, “Arousal”, and “Dominance” separately. “ACC” means accuracy and “F1” indicates the F1-score. The *bold* values indicate the better results.)

Model	Emo-MG (%)			IEMOCAP (%)		EMOTIC (%)		
	V(ACC/F1)	A(ACC/F1)	D(ACC/F1)	Accuracy	F1-Score	V(ACC/F1)	A(ACC/F1)	D(ACC/F1)
SVM	77.22/ 67.29	67.09/ 61.09	62.03/ 60.59	68.65	68.08	72.59/ 64.27	62.30/ 62.31	81.23/ 71.63
KNN	78.48/ 69.02	65.82/ 62.49	65.82/ 62.66	63.22	62.99	69.74/ 64.26	61.93/ 61.82	80.98/ 70.42
RF	81.01/ 73.65	67.08/ 65.08	64.56/ 64.04	70.45	69.71	72.75/ 65.00	61.84/ 61.43	80.04/ 71.18
SGD	79.75/ 73.01	62.03/ 59.61	60.76/ 57.17	59.76	60.01	62.68/ 62.08	58.01/ 58.11	71.49/ 71.48
NBC	73.42/ 72.73	68.35/ 59.47	60.76/ 58.56	59.37	60.08	52.18/ 54.09	62.59/ 62.68	55.02/ 59.70
AdaBoost	75.95/ 71.18	64.56/ 63.72	63.29/ 60.88	65.28	65.15	71.02/ 62.14	61.52/ 61.37	76.49/ 70.15
MLP	81.01/ 75.61	69.62/ 69.62	69.62/ 67.17	62.93	63.26	71.41/ 61.61	61.18/ 59.47	81.36/ 71.37
KANFeel	81.81/ 75.84	72.48/ 75.28	70.15/ 68.96	70.64	70.37	71.70/ 62.05	62.82/ 62.89	80.81/ 72.21

Table 3: MER Performances Comparison with Deep Learning and SOTA Methods (noted by *)

Model	Emo-MG (%)			Model	IEMOCAP (%)		Model	EMOTIC (%)		
	V(ACC/F1)	A(ACC/F1)	D(ACC/F1)		Accuracy	F1-Score		V(ACC/F1)	A(ACC/F1)	D(ACC/F1)
RGNN*	82.14/ 74.69	72.15/ 65.83	68.35/ 63.89	CORECT*	74.39	73.50	EMOT-Net*	72.43/ 64.98	64.27/ 63.99	81.31/ 73.41
Emo-MG-LSTM*	82.28/ 74.28	75.95/ 75.81	69.62/ 67.84	COGMEN*	72.09	71.50	GNN-CNN*	73.18/ 65.21	64.52/ 64.14	81.20/ 74.23
KANFeel	81.81/ 75.84	72.48/ 75.28	70.15/ 68.96	KANFeel	70.64	70.37	KANFeel	71.70/ 62.05	62.82/ 62.89	80.81/ 72.21
KANFeel-Attent	83.54/ 78.85	77.68/ 76.03	73.42/ 70.85	KANFeel-Attent	72.16	71.58	KANFeel-Attent	73.41/ 65.49	64.96/ 64.38	81.80/ 74.93

5.2 Effectiveness of KAN-based Model

We first summarized all experimental results in Table 2 by comparing KAN-based with other baseline ML models. In summary, KANFeel consistently outperforms the other models across all datasets and emotional dimensions, achieving the highest accuracy and F1-scores in most scenarios. On Emo-MG, KANFeel outperforms other models with the best accuracy and F1-score results across all three VAD dimensions. Besides, RF and MLP-based models also indicated significant performances. On EMOTIC, the RF-based model achieves the best performances in the “Valence” dimension, while KANFeel still delivers the highest scores in the other two dimensions. On IEMOCAP with discrete emotional labels, KANFeel also shows superior performances in MER tasks considering both accuracy and F1-score results.

Besides, a comprehensive comparison with deep learning and SOTA methods is summarized in Table 3. Across all three datasets, the effectiveness of both **KANFeel** and **KANFeel-Attent** is evident. The integration of the attention mechanism in **KANFeel-Attent** leads to significant improvements over the standard KANFeel model. Notably, **KANFeel-Attent** demonstrates outstanding performance across all datasets, surpassing other deep learning and SOTA methods on the Emo-MG and EMOTIC datasets. While **KANFeel-Attent** trails CORECT on the IEMOCAP dataset, it still outperforms the COGMEN model in MER tasks.

6 Conclusion and Future Work

Empowered by the Kolmogorov-Arnold Networks (KANs), this paper presents a novel framework, **KANFeel**, for multimodal emotion recognition (MER) tasks. In our experiments, we evaluated the performance of various machine learning methods and state-of-the-art (SOTA) approaches across three multimodal emotion datasets. Our results confirm two key findings in MER tasks: 1) Compared to MLP, KAN demonstrates superior efficiency in terms of model parameters and training costs. 2) KAN exhibits strong effectiveness in MER and shows significant potential for further enhancement when integrated with other advancing models, such as **KANFeel-Attent**, which incorporates KAN-based attention mechanisms.

From a user-centric perspective, **KANFeel** offers several advantages that can improve user experience in real-world applications. First, the efficiency of KAN in terms of model parameters and training costs can facilitate ease of integration into resource-constrained environments, such as mobile devices or edge computing systems [45]. Furthermore, the **KANFeel-Attent** framework, which incorporates KAN-based attention mechanisms, has the potential to provide more personalized and context-aware emotional insights due to the outstanding multimodal analysis ability [25], further improving both usability and perceived relevance to users.

For future work, several promising directions can be explored. One avenue is the exploration of more advanced deep learning models with KANs, such as large language models (LLMs) [46], which

could offer improved performance by capturing richer dependencies within multimodal data. Besides, fine-tuning and optimizing the attention mechanisms based on **KANFeel-Attent** could lead to even greater performance gains compared to the SOTA methods in most cases, making the framework more adaptable for more transformer-based models in various MER applications. Finally, an interactive platform can be deployed with various MER models, and more subjective evaluations can be conducted to investigate users' attitudes from both functionality and usability angles. Further challenges should be further explored in practical implementation scenarios, including achieving low-latency processing in real-time, effectively collecting and integrating heterogeneous data across multiple modalities from users, ensuring ethical considerations and maintaining user trust [2].

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